Mock 05

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1 INSTRUCTIONS

- **Time Allowed:** 120 Minutes.
- **Total Marks:** 41 (Scaled to 100).
- **Format:**
  - **Section A:** 10 Multiple Choice Questions (1 point each).
  - **Section B:** 10 Short Open Problems (1.5 points each).
  - **Section C:** 8 Complex Open Problems (2 points each).
- **Constraint:** No calculators allowed. Show all rigorous steps in Sections B and C.

2 Section A: Multiple Choice (10 Points)

Select the single best answer.

2.1 Problem A1

Which of the following statements is the **contrapositive** of: "If a sequence is convergent, then it is bounded"?

(a) If a sequence is bounded, then it is convergent.

(b) If a sequence is not bounded, then it is not convergent.

(c) If a sequence is not convergent, then it is not bounded.

(d) A sequence is convergent if and only if it is bounded.

2.2 Problem A2

Let $S = \{a, b, c\}$. What is the cardinality of the power set $\mathcal{P}(S)$?

(a) 3 (b) 6 (c) 8 (d) 9

2.3 Problem A3

Compute the value of i^{2026}.

(a) 1 (b) -1 (c) i (d) $-i$

2.4 Problem A4

Consider the following Julia code. What is the output?

```
A = [1, 2, 3, 4]
B = [x^2 for x in A if x % 2 == 0]
println(sum(B))
```

(a) 10 (b) 20 (c) 30 (d) 4

2.5 Problem A5

Which of the following congruences has **no** integer solution?

(a) $x^2 \equiv 1 \pmod 5$ (b) $2x \equiv 3 \pmod 7$ (c) $x^2 \equiv 3 \pmod 7$ (d) $3x \equiv 6 \pmod 9$

2.6 Problem A6

Evaluate the limit: $\lim_{n\to\infty} \frac{4n^2+\sin(n)}{2n^2-3}$.

(a) 0 (b) 2 (c) ∞ (d) 4

2.7 Problem A7

Let R be a relation on $\mathbb{Z}$ defined by $xRy \iff x \leq y$. Which property does R **fail**?

(a) Reflexive (b) Anti-symmetric (c) Transitive (d) Symmetric

2.8 Problem A8

What is the coefficient of x^2y^2 in the expansion of $(x+2y)^4$?

(a) 4 (b) 6 (c) 12 (d) 24

2.9 Problem A9

A function $f : A \to B$ is **surjective** if:

(a) $|A| < |B|$ (b) $f(A) = B$ (c) $f(x) = f(y) \implies x = y$ (d) $f(A) \subseteq B$

2.10 Problem A10

If vectors $\mathbf{u}$ and $\mathbf{v}$ satisfy $|\mathbf{u}| = 2, |\mathbf{v}| = 3$, and the angle between them is $\pi/3$, what is $\mathbf{u} \cdot \mathbf{v}$?

(a) 3 (b) 6 (c) $3\sqrt{3}$ (d) 0

3 Section B: Short Open Problems (15 Points)

Provide a concise solution and final answer.

3.1 Problem B1

Write a Julia function ‘sum_{odds}(n)‘ that returns the sum of all odd integers from 1 to n (inclusive).

3.2 Problem B2

Simplify the logical expression $\neg(P \vee (\neg P \wedge Q))$.

3.3 Problem B3

Calculate the modulus of the complex number $z = \frac{3+4i}{1-i}$.

3.4 Problem B4

Let $A = [0, 3]$ and $B = (1, 4)$ be subsets of $\mathbb{R}$. Find the symmetric difference $A \triangle B$.

3.5 Problem B5

How many distinct permutations can be made from the letters of the word “MAMA”?

3.6 Problem B6

Find the smallest positive integer x such that $4x \equiv 5 \pmod 9$.

3.7 Problem B7

Find the remainder when the polynomial $P(x) = x^{100} + 2x^2 + 1$ is divided by $x - 1$.

3.8 Problem B8

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{x}{x+2}$, where domain is $\mathbb{R} \setminus \{-2\}$.

3.9 Problem B9

Calculate the projection of vector $\mathbf{a} = (2, 3)$ onto vector $\mathbf{b} = (4, 0)$.

3.10 Problem B10

Evaluate the limit: $\lim_{n\to\infty}(\sqrt{n^2 + 2n} - n)$.

4 Section C: Complex Open Problems (16 Points)

Provide rigorous proofs. State all lemmas or theorems used.

4.1 Problem C1 (Induction)

Prove by Mathematical Induction that for all integers $n \geq 1$:

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

4.2 Problem C2 (Relations)

Let $\sim$ be a relation on $\mathbb{Z}$ defined by $x \sim y \iff x + y$ is even.

1. Prove that $\sim$ is an equivalence relation.
2. Describe the distinct equivalence classes.

4.3 Problem C3 (Analysis)

Using the formal $\epsilon - N$ definition of the limit, prove that:

$$\lim_{n\to\infty} \frac{2n}{n+1} = 2$$

4.4 Problem C4 (Set Theory)

Prove or disprove: For any sets A, B, C,

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

4.5 Problem C5 (Combinatorics)

Five points are placed inside an equilateral triangle of side length 2. Prove that there exist at least two points whose distance is less than or equal to 1. (Hint: Pigeonhole Principle).

4.6 Problem C6 (Complex Analysis)

Find all complex roots of the equation $z^3 = 8i$. Express the roots in the form $a + bi$.

4.7 Problem C7 (Cardinality)

Prove that the set of all integers $\mathbb{Z}$ is countable by constructing an explicit bijection $f : \mathbb{N} \to \mathbb{Z}$.

4.8 Problem C8 (Number Theory)

Solve the following system of congruences:

$$\begin{cases} x \equiv 1 \pmod{3} \\ x \equiv 2 \pmod{5} \\ x \equiv 3 \pmod{7} \end{cases}$$

5 Section D: Answer Key & Grading Rubric

5.1 Section A (MCQ)

Q	Ans	Topic
1	B	Logic (Contrapositive)
2	C	Sets ($2^3 = 8$)
3	B	Complex ($2026 \equiv 2 \pmod 4$)
4	B	Julia ($2^2 + 4^2 = 4 + 16 = 20$)
5	C	Quadratic Residues (3 is not a QR mod 7)
6	B	Limits (Ratio of leading coeffs)
7	D	Relations (Order is not symmetric)
8	D	Binomial Thm ($\binom{4}{2} \cdot 1^2 \cdot 2^2 = 6 \cdot 4 = 24$)
9	B	Functions (Definition of Surjective)
10	A	Vectors ($\lvert\mathbf{u}\rvert\lvert\mathbf{v}\rvert \cos\theta = 2 \cdot 3 \cdot \frac{1}{2} = 3$)

5.2 Section B (Short Open)

B1: ‘function sum_{odds}(n); return sum(1:2:n); end‘ (or loop). **[1.5 pts]**

B2: $\neg P \wedge \neg(\neg P \wedge Q) \equiv \neg P \wedge (P \vee \neg Q) \equiv (\neg P \wedge P) \vee (\neg P \wedge \neg Q) \equiv \neg P \wedge \neg Q$. **[1.5 pts]**

B3: $|3 + 4i|/|1 - i| = 5/\sqrt{2} = 2.5\sqrt{2}$. **[1.5 pts]**

B4: $A \cup B = [0, 4)$, $A \cap B = (1, 3]$. Diff $= [0, 1] \cup (3, 4)$. **[1.5 pts]**

B5: $\frac{4!}{2!2!} = \frac{24}{4} = 6$. **[1.5 pts]**

B6: Inverse of 4 mod 9 is 7. $x \equiv 5(7) \equiv 35 \equiv 8 \pmod 9$. **[1.5 pts]**

B7: $P(1) = 1^{100} + 2(1)^2 + 1 = 4$. **[1.5 pts]**

B8: $y = x/(x + 2) \Rightarrow xy + 2y = x \Rightarrow x(y - 1) = -2y \Rightarrow x = 2y/(1 - y)$. **[1.5 pts]**

B9: $\text{proj}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|^2}\mathbf{b} = \frac{8}{16}(4, 0) = (2, 0)$. **[1.5 pts]**

B10: Multiply conjugate: $\frac{2n}{\sqrt{n^2+2n}+n} \to \frac{2n}{2n} = 1$. **[1.5 pts]**

5.3 Section C (Complex Open)

C1 (Induction)

- **Base:** $n = 1 \Rightarrow 1 = \frac{1(2)(3)}{6} = 1$.
- **Hypothesis:** Assume $\sum_1^k = \frac{k(k+1)(2k+1)}{6}$.

- **Step:** Add $(k+1)^2$. Factor $\frac{k+1}{6}(k(2k+1)+6(k+1)) = \frac{k+1}{6}(2k^2+7k+6)$.
- Factor quadratic: $(2k+3)(k+2)$. Result matches $n = k+1$. **[2 pts]**

C2 (Relations)

- **Reflexive:** $x + x = 2x$ (even).
- **Symmetric:** $x + y$ even $\implies y + x$ even.
- **Transitive:** $x + y = 2k, y + z = 2j$. Sum: $x + 2y + z = 2(k + j) \implies x + z = 2(k + j - y)$ (even).
- **Classes:** Evens [0] and Odds [1]. **[2 pts]**

C3 (Limit Proof)

- $|\frac{2n}{n+1} - 2| = |\frac{2n-2n-2}{n+1}| = \frac{2}{n+1}$.
- We need $\frac{2}{n+1} < \epsilon$.
- Choose $N > \frac{2}{\epsilon} - 1$.
- For all $n > N$, inequality holds. **[2 pts]**

C4 (Sets)

- **True.** De Morgan's Laws for sets.
- $x \in A \setminus (B \cap C) \iff x \in A \wedge x \notin (B \cap C)$.
- $\iff x \in A \wedge (x \notin B \vee x \notin C)$.
- $\iff (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C)$.
- $\iff x \in (A \setminus B) \cup (A \setminus C)$. **[2 pts]**

C5 (Combinatorics)

- Divide triangle into 4 smaller equilateral triangles of side length 1 (connect midpoints).
- We have 5 points and 4 regions.
- By PHP, at least one region contains 2 points.
- Max distance in equilateral triangle of side 1 is 1. **[2 pts]**

C6 (Complex)

- $z^3 = 8e^{i(\pi/2+2\pi k)}$.
- Roots: $2e^{i(\pi/6+2\pi k/3)}$.
- $k = 0 : 2(\cos 30^\circ + i \sin 30^\circ) = \sqrt{3} + i$.
- $k = 1 : 2(\cos 150^\circ + i \sin 150^\circ) = -\sqrt{3} + i$.
- $k = 2 : 2(\cos 270^\circ + i \sin 270^\circ) = -2i$. **[2 pts]**

C7 (Cardinality)

- Map $f(n)$ as follows:
- Even $n = 2k \to k$ (Positive).
- Odd $n = 2k + 1 \to -k$ (Negative).
- $f(0) = 0, f(1) = -1, f(2) = 1, f(3) = -2 \ldots$ (Zig-zag mapping).
- This is a bijection. **[2 pts]**

C8 (Number Theory)

- $x \equiv 1 \pmod 3 \wedge x \equiv 2 \pmod 5 \implies x = 3k+1 \implies 3k+1 \equiv 2(5) \implies 3k \equiv 1 \equiv 6(5) \implies k \equiv 2(5)$.
- $x = 3(5j + 2) + 1 = 15j + 7$.
- $15j + 7 \equiv 3 \pmod 7 \implies 1j + 0 \equiv 3(7) \implies j \equiv 3(7)$.
- $x = 15(7m + 3) + 7 = 105m + 45 + 7 = 105m + 52$.
- Solution: 52. **[2 pts]**